AFE

Project Statement

Consolidated and amended from the House Agent / HomeConnect Africa document set

A trust-first property platform for African renters, buyers, landlords and agents. Ghana first.

Field	Value
Product	Afe (working names retired: House Agent, HomeConnect Africa, Sheltr)
Document	Project Statement — single source of truth for scope, architecture and plan
Version	3.3 (draft) — v3.2 plus the business model and viability section (Section 12). Earlier: v3.1 stack change STACK-001, v3.2 storage and backup
Date	15 September 2026
Owner	Raphael
Supersedes in part	Engineering Docs v2.0, Frontend Gap Implementation Plan v1.0, ROADMAP.md, naming-options.md. Other files remain as detailed references (see Appendix B).

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How to read this document

Sections 1–6 cover what Afe is, who it serves and what ships first. Sections 7–10 cover compliance, architecture, AI and metrics. Sections 11–15 cover the plan, the business model, team, risks and open decisions.

[DECISION] marks a point the founder must confirm before engineering starts. Every change from the source documents is logged in Appendix A.

Legal note

Legal and regulatory points here are summaries for planning. They are not legal advice. Confirm each one with a Ghana Bar Association–licensed lawyer (and local counsel in each later market) before launch.

1. Executive summary

Afe is a property platform where every listing is checked before it goes live, every agent is verified, and rent money moves through escrow instead of cash in hand. It launches in Ghana, starting with Accra and Kumasi.

Renters stop paying viewing fees for houses that don't exist or that the "agent" has no right to let. Honest landlords and agents get a way to prove they are honest, and they get found faster because of it.

The source documents describe a much larger system: four countries at once, eleven microservices, six AI model pods, 100,000 concurrent users and $25K MRR by week 34. This statement keeps that as the long-term direction. It cuts Release 1 down to what a small team can build, test and legally operate in one country.

Item	Position in this statement
Launch market	Ghana — Accra and Kumasi
Users at launch	Tenants/buyers, landlords/owners, agents, Afe admins
Core promise	Verified listings, verified agents, escrowed deposits and advance rent
Backend	Python (FastAPI) modular monolith, with Rust modules for photo hashing, image processing and scoring; PostgreSQL 16 + PostGIS; Redis; Typesense
Clients	SvelteKit web app (PWA); native Android app (Kotlin + Jetpack Compose); native iOS app (Swift + SwiftUI)
Payments	Mobile money (MTN MoMo, Telecel Cash, AT Money) through a licensed aggregator
Expansion	Nigeria and Kenya next, South Africa after; Tanzania moved to backlog
Target Ghana launch	About week 28 from kick-off; Android beta first, iOS by launch (see Section 11)

2. Problem statement

Looking for a home in Ghana usually means dealing with informal agents, WhatsApp photos and cash payments. The same pattern shows up in Nigeria and Kenya. The main failures are these:

- **Fake and duplicate listings.** Photos are copied from other sites, and the same house is posted by several people who claim to represent it.
- **Fees for nothing.** Renters pay viewing or "inspection" fees to people with no right to let the property.
- **Illegal upfront rent.** Landlords commonly ask for one or two years in advance. Ghana's Rent Act, 1963 (Act 220) caps advance rent at six months for tenancies longer than six months, and the Rent Control Department said in 2026 that it would prosecute breaches from 1 April 2026.
- **Unlicensed agents.** The Real Estate Agency Act, 2020 (Act 1047) requires agents to be licensed, but a renter has no quick way to check.
- **No safe way to pay.** Deposits go by cash or direct mobile money to a stranger. If the deal falls through, there is no recourse.
- **Little market data.** Renters and buyers can't easily tell whether a price is fair for the area.

Afe's answer is to make trust the product: verify the property, verify the person, and hold the money until both sides confirm.

3. Vision and product principles

Vision. Become the most trusted place in Africa to find, rent and buy a home.

Mission for Release 1. Make it safe for a renter in Accra or Kumasi to find a real home, meet a real agent and pay a legal deposit without being scammed.

3.1 Principles

Principle	What it means in practice
Trust before growth	No listing goes live without a verification score. Fewer, real listings beat many fake ones.
Mobile first, low data	Most users are on phones and paid data. Aim for an initial web load under 100 KB, WebP images, lazy loading and offline-friendly PWA behaviour.
The law is configuration	Rent caps, notice periods and similar rules live in versioned config, not in code, so a legal change doesn't need a deployment.
Every flag explains itself	When Afe blocks or flags something, the user and the reviewer both see the reasons.
Build one thing, split later	Start as one well-structured service. Split modules out only when load or team size demands it.
Humans decide the hard cases	Automation scores and sorts. People make final calls on disputes and borderline fraud until there is data to do better.

4. Users and roles

Role	What they need	Key Afe features
Tenant / buyer	Real listings, fair prices, safe payment, a way to complain	Search and map, verified badges, viewing requests, GPS-gated reviews, escrow, affordability calculator, saved searches
Landlord / owner	Serious tenants, protection from agent fraud, on-time payment	KYC, listing management, ownership proof upload, escrow payouts, tenant enquiries, basic performance stats
Agent	Leads, a credible profile, less time wasted	Licence verification, public profile, listing pipeline, viewing calendar, messaging
Afe admin	Stop fraud, resolve disputes, stay compliant	Verification and fraud queues, dispute console, payments view, compliance rules editor (dual approval), audit log
Lender (later)	Qualified borrowers	Loan matching and pre-approval (Release 3)
Developer partner (later)	Data and integration	Public API and webhooks (Release 3)

5. Scope

5.1 Release 1 — Ghana MVP (in scope)

Module	What ships in Release 1
Auth & identity	Phone or email sign-up with OTP; JWT access + refresh tokens; TOTP MFA required for landlords, agents and admins; Ghana Card check through an NIA-backed provider; device list and logout-all.
Listings	Create and edit, photo upload, map pin plus GhanaPostGPS digital address, status flow: draft → pending review → live → let/sold → archived.
Verification	0–100 verification score, duplicate-photo detection with perceptual hashing, ownership or landlord-authorisation document upload, agent licence check, manual review queue.
Search	Typesense text search with typo tolerance, radius and map filters, facets (price, bedrooms, verified only), saved searches with alerts.
Viewings & reviews	Request and confirm viewings, GPS check-in on site, reviews allowed only with a valid check-in token.
Escrow payments	Deposit and advance rent via mobile money, escrow state machine, 72-hour dispute window, Act 220 caps enforced at payment time, receipts.
Messaging & notifications	In-app threads between tenant and agent or landlord over WebSocket; SMS, email and push notifications.
Dashboards	Tenant, landlord, agent and admin dashboards, reduced to the cards needed for the flows above.
Calculators	Rent affordability and simple mortgage calculator, computed client-side. Moved forward from Phase 3 because it is cheap to build.
Localisation	English (en-GH), GHS, Africa/Accra time zone. Twi strings for the main flows are a stretch goal **[DECISION]**.

5.2 Deferred features

The source documents list eleven "differentiated features" and seven work-item groups. They stay on the roadmap, re-sequenced by what they depend on.

Feature	Release	Why it waits
ML price model (XGBoost) and recommendations (LightFM)	R2	Needs months of Ghana listing and interaction data. Rules and filters cover R1.
Image classifier (ResNet) for stock/fake photos	R2	Perceptual hashing plus manual review catches most copies first; the classifier needs labelled examples.
LSTM GPS trajectory model	R2	Needs labelled real visits. R1 uses rule-based motion checks.
Market and neighbourhood intelligence	R2 basic stats, R3 predictions	Needs listing and transaction volume per area.
Seller optimiser (pricing, photo quality, timing)	R2	Depends on the price model.
Reputation system with multi-party trust scores	R2	R1 ships verified badges and review counts; full scoring needs history.

Feature	Release	Why it waits
Loan matching and lender integration	R3	Needs signed lender partners. Marked CRITICAL in the Implementation Plan, but nothing can ship without partners.
AI dispute mediation	R3	R1 disputes are handled by people; their outcomes become training data.
Developer API and webhooks	R3	Internal API must stabilise first.
Nigeria and Kenya	R3	Ghana must prove the model first.
South Africa	R4	Needs an interest-bearing deposit custodian and FICA process.
AR / 360° tours, predictive maintenance, voice search	R4	Nice to have; high cost.
Causal inference (DoWhy / EconML)	R4	Research-grade; only useful with large data.
Tanzania	Backlog	Present in early docs, absent from later ones.

5.3 Out of scope for now

- Holding customer money in an Afe-owned account (see 6.3).
- Full property-management accounting for landlords.
- Short-let and holiday bookings.
- Commercial property leases.

6. Trust and safety design

Trust is the product, so this section is the heart of the build.

6.1 Listing verification score

Each listing gets a 0–100 score at creation and again when key fields change. The weights come from fraud-detection.md, with one Release 1 amendment.

Component	Source weight	R1 weight	Notes
Ownership documents	30%	30%	Many Ghana rentals have no title deed. Accept a Lands Commission record, indenture or site plan, or a signed landlord authorisation letter for agents.
Agent / landlord reputation	25%	30%	Includes licence status, Ghana Card match, account age and dispute history.
Property metadata	20%	20%	Completeness, GPS pin matches address, photo count and duplicates.
Market analysis	15%	20%	Price far below the area range (below 60% of the low end) is a scam signal.
ML anomaly score	10%	0% → 10% in R2	No trained model at launch. Weight moves to the other parts until one exists.

Risk bands stay as written: 80+ low, 60–79 medium, 40–59 high, below 40 critical. Medium and below go to the manual review queue; critical listings are blocked.

6.2 GPS check-in

A check-in proves someone was physically at the property. It gates reviews and certain payment steps. It does not prove the person is allowed to let the property; that comes from 6.1.

Layer	RFC-001 (source)	Release 1 (amended)
1. Device integrity	Play Integrity (Android), DeviceCheck (iOS); reject rooted devices	Same, plus iOS App Attest. Rooted devices are sent to manual review rather than blocked outright, so honest users on older phones aren't locked out.
2. Motion & sensors	Accelerometer + gyroscope during a 60-second window	Same, plus the Android mock-location flag and GPS accuracy radius.
3. Anomaly score	LSTM on visit trajectories	Rule-based score in R1 using the same thresholds: below 0.60 reject, 0.60–0.79 review, 0.80+ approve. LSTM in R2.

The check-in score is computed in Rust (`afe_native.checkin_score`) inside the Python API. On success Afe issues a signed check-in token (24-hour TTL), stored in Redis. Every attempt, pass or fail, goes to the audit log.

6.3 Escrow

State	Meaning	Next
INITIATED	Tenant has paid; funds sit with the payment partner, not in an Afe account	LANDLORD_CONFIRMED or DISPUTED

State	Meaning	Next
LANDLORD_CONFIRMED	Landlord confirms move-in; auto-advances after 24 h if no dispute	RELEASED
DISPUTED	Either party raises an issue; both notified; 72-hour window	RESOLVED_TENANT or RESOLVED_LANDLORD
RESOLVED_TENANT	Decision for tenant	REFUNDED
RESOLVED_LANDLORD	Decision for landlord	RELEASED
RELEASED	Paid to landlord minus platform fee	End
REFUNDED	Returned to tenant minus gateway fee	End

Rules enforced at INITIATED for Ghana: no more than six months' advance rent for tenancies longer than six months, and no more than two months for shorter tenancies. Some guides say renewals are limited to three months' advance; confirm this before encoding it. Idempotency keys are required on every payment call.

Escrow licensing [DECISION]

Holding money for others is regulated. Before launch, get a written legal opinion on the Payment Systems and Services Act, 2019 (Act 987) and Bank of Ghana rules, and choose a licensed partner (for example Hubtel or Paystack) whose product supports held or split settlements.

The platform fee (2.5% in RFC-002) and who pays it — tenant, landlord or split — also needs a decision.

6.4 Fraud detection pipeline

- **On listing create (synchronous):** perceptual hash of each photo against the index, verification score, price check. Hashing and scoring run in Rust; Python keeps the weights and the database writes. Target under 300 ms p95 excluding image download.
- **Background (Celery):** Isolation Forest on user behaviour (sign-up bursts, many listings from one device, message patterns).
- **Nightly (R2):** graph job linking accounts, devices, phone numbers and payout wallets to find fraud rings.
- **Human loop:** every reviewer decision is stored as a label for later models.

Measure fraud detection with precision and recall on a labelled test set, not with plain accuracy. Fraud is rare, so a model that flags nothing can still score 99% accuracy.

7. Markets and compliance

Rules below replace the country table in RFC-004 and the regional sections of README.md. Items marked "confirm" were not verified for this statement.

Country	Payments	Key laws and checks	Languages
Ghana (R1)	GHS. MTN MoMo, Telecel Cash (formerly Vodafone Cash), AT Money via Hubtel and/or Paystack	Rent Act, 1963 (Act 220): six-month advance cap. Real Estate Agency Act, 2020 (Act 1047): agent licensing. Data Protection Act, 2012 (Act 843): register with the Data Protection Commission. Ghana Card (NIA) for identity. A new Rent Bill has been before Parliament since 2023; track it.	English; Twi, Ga, Ewe for search and key flows
Nigeria (R3)	NGN. Paystack, Flutterwave	Nigeria Data Protection Act, 2023 (replaces the NDPR reference). NIN verification. State tenancy laws; Lagos stamp duty.	English, Yoruba, Igbo, Hausa
Kenya (R3)	KES. M-Pesa (Daraja API)	Data Protection Act, 2019. Deposit cap of two months in RFC-004 — confirm, as the source is unclear.	English, Swahili
South Africa (R4)	ZAR. PayFast, Peach Payments	POPIA. Rental Housing Act, 1999: deposit held in an interest-bearing account. PIE Act for evictions. FICA KYC.	English, isiZulu (others later)

7.1 Compliance engine

RFC-004 stays as the design, with in-process calls instead of gRPC for Release 1:

- Rules stored as versioned rows: country, region, rule type, value, effective date, source law, version.
- Cached in Redis and refreshed at most every 15 minutes.
- Changed only through the admin dashboard with two named legal approvers, with one-click rollback.
- Every rule check written to an append-only audit table.

7.2 Payment card security

The Engineering Docs target a "PCI-DSS audit in all four countries". PCI-DSS is a card-industry standard, not a national one. Afe should never touch raw card data: use the gateway's hosted checkout so Afe stays in the lightest PCI scope. Mobile money is the main rail anyway.

8. Technical architecture

8.1 Resolved decisions

The source files contradict each other on several core choices. These are the calls this statement makes.

Area	Decision	Conflict it resolves	Reason
Backend shape	Python (FastAPI) modular monolith with 12 modules and strict boundaries	"8" and "11" microservices in Engineering Docs (the 11-service table lists 12)	framework-comparison.md, docker-compose.yml and kubernetes.yaml all describe one API. A small team can't run 12 services.
Web app	SvelteKit 2, Svelte 5 runes, TypeScript strict, Tailwind	React 18 (Eng Docs O2), Vue (early prompts), Next.js + shadcn (Gray CSM)	Quick Fixes Fix 01 already chose SvelteKit.
Performance-critical code	Rust, compiled into a Python module (`afe_native`) with PyO3 and maturin; Rust Axum services only from R2 if load tests need them	"Move hot paths to Go" (framework-comparison.md)	Founder decision (STACK-001). Pure functions only; Python owns business logic and data.
Android	Kotlin + Jetpack Compose, Android 8.0+	Flutter 3 (single codebase)	Founder decision (STACK-001). Direct access to Play Integrity, sensors and location; faster start-up on low-end phones.
iOS	Swift + SwiftUI, iOS 16+ **[DECISION D13]**	Flutter 3 (iOS 14+)	Founder decision (STACK-001). Direct App Attest, Core Location and Core Motion.
Mobile contracts	Kotlin and Swift API clients generated from the FastAPI OpenAPI spec; shared design-token file	Hand-written Dart models	Keeps two native apps in step with the web app.
Admin UI	Admin role inside the SvelteKit app	Django Admin (SKILL.md)	Avoids a second backend framework; admin dashboard prompts already exist.
Database	PostgreSQL 16 + PostGIS + pg_trgm; row-level security by country	Postgres 15 in docker-compose vs 16 in Eng Docs	Pick one version everywhere.
Search	Typesense	Elasticsearch (Eng Docs, PR-002)	Already in docker-compose; much lighter to run.
Jobs and events	Celery + Redis; Redis Streams for domain events	Kafka	Kafka is heavy to operate. Revisit when a module is split out.
Internal calls	Python interfaces between modules	gRPC	No network hop inside one process.
Real-time	FastAPI WebSockets; the 11 topics from the API InfraPrompt	None	Keep topic names as specified.

Area	Decision	Conflict it resolves	Reason
Hosting	AWS, containers. ECS Fargate or small EKS **[DECISION]**	EKS from day one	Fargate is simpler for R1.
Region	**[DECISION]** af-south-1 (Cape Town) or eu-west-1 (Ireland)	Docs say eu-west-1 primary; also mention Lagos/Nairobi replicas	Weigh latency to Ghana and Act 843 cross-border transfer rules.
File storage	Amazon S3, five buckets split by sensitivity; photos served through CloudFront (Section 8.6)	One "S3 Object Storage" box; image URLs stored in the database	KYC files need far tighter access than listing photos.
Backup	RDS point-in-time recovery + AWS Backup copies to a second region and a separate AWS account (Section 8.7)	30-day RDS snapshots only	Meets RPO ≤ 1 h and survives a compromised main account.
CI/CD	GitHub Actions; ArgoCD later	None	Keep the under-8-minute build target.
Secrets	AWS Secrets Manager; no secrets in code	None	From O1.
Observability	Prometheus, Grafana, JSON logs, OpenTelemetry	Datadog vs ELK undecided	Open-source stack is already in docker-compose.
Model tracking	MLflow (recommended) **[DECISION]**	MLflow vs DVC (RFC-003)	Tracks runs and serves registry in one tool.

8.2 Module map

The modules match the service list in the API InfraPrompt so the frontend service clients don't change. Any module can later become its own service.

Module	Base path	Release 1 scope
auth	/api/v1/auth	Full
properties	/api/v1/properties	Search, detail, reviews, saved, alerts
verify (GPS)	/api/v1/verify	Check-in, token validation
kyc	/api/v1/kyc	Ghana Card, documents, agent licence
payments	/api/v1/payments	Escrow flow, webhooks, transactions
compliance	/api/v1/compliance	Ghana rules, validate, tenancy agreement template
notifications	/api/v1/notifications	SMS, email, push, in-app list
messages	/api/v1/messages	Threads and send; AI assist in R2
agents	/api/v1/agents	Directory, profile, viewings; optimiser in R2
ai	/api/v1/ai	Fraud score and explanation only; match/price in R2
intelligence	/api/v1/intelligence	Not in R1; basic area stats in R2
finance	/api/v1/finance	Calculator endpoints only; loans in R3

8.3 API conventions

- REST under /api/v1, response envelope { data, meta, error }.
- Headers: Authorization (Bearer), X-Country-ID, X-Correlation-ID, and Idempotency-Key on every payment mutation.
- Cursor pagination for feeds and search; offset pagination for admin tables.
- Timeouts, retries and debounce values as set in Quick Fixes Fix 08.
- Frontend: native fetch client, silent token refresh, MSW mocks in development, Vitest for tests (per implementation_plan.md).

8.4 Rust and native mobile

Full detail is in the Stack Change Amendment (STACK-001): which functions move to Rust, the crate layout, the Docker build, the Android and iOS architecture, and replacement prompts for every Flutter component. The rule in short: Rust only for pure, CPU-heavy functions with shared test fixtures; everything with business rules or database writes stays in Python.

8.5 Data model gaps

database-schema.md says 26 tables but defines about 15. Release 1 also needs these, added through Alembic migrations:

- agents and agent_licences
- viewings and gps_checkins
- reviews (linked to a check-in token)
- message_threads and messages
- saved_searches and alerts
- kyc_documents and ownership_documents
- escrow_disputes
- compliance_rules (versioned) and compliance_evaluations
- notifications
- stored_files (one row per S3 object: bucket, key, hash, owner, sensitivity, scan status)
- outbox_events (payment and escrow events written here before they are published)

property_images keeps its hashes and moderation fields but stores an S3 key instead of a full URL.

8.6 File storage

All files live in Amazon S3. PostgreSQL stores only each file's bucket, key, hash and metadata, never the file itself. Buckets are split by how sensitive the contents are.

Bucket	Holds	Who can read it	Protection
afe-media-public	Processed listing photos (WebP, several sizes)	Everyone, only through CloudFront	Can be rebuilt from originals; no replication
afe-media-originals	Raw uploaded photos; chat photos (private prefix)	Backend only; signed URLs for chat	Versioning; cross-region replication; kept as fraud evidence
afe-kyc-private	Ghana Card images, ownership documents, landlord authorisation letters, agent licences	Admin reviewers only, through signed URLs that expire in minutes	Customer-managed KMS key; public access blocked; every read logged; replication

Bucket	Holds	Who can read it	Protection
afe-documents	Tenancy agreements, escrow receipts, dispute evidence	The parties to the deal and admins	Versioning; Object Lock so records can't be altered or deleted early; replication
afe-backups	Database snapshots and exports	No application access	Separate AWS account; Object Lock; second region

Upload flow

1. The app asks the API for a presigned upload URL. The file goes straight to S3, not through the API server.
2. The file lands in a quarantine prefix. An S3 event starts a worker.
3. The worker checks the real file type and size, and runs a malware scan.
4. For photos, it reads the GPS and date metadata into the database for verification, then strips all metadata so a landlord's location is never published.
5. It hashes the photo and builds WebP sizes with the Rust module (afe_native), then moves the file to its final bucket.
6. Rejected files stay in quarantine for 7 days, then are deleted.

Signed download URLs for KYC files expire within 5 minutes and are issued only after an admin permission check. Chat photos use signed URLs that expire within an hour.

Lifecycle and cost

- Original photos move to a cheaper storage class after about 90 days.
- Photos from deleted listings and KYC files for closed accounts are kept only as long as the retention policy allows **[DECISION D18]**.
- Storage is cheap at MVP scale; CloudFront image delivery is the larger cost. Cloudflare R2 is an option for public photos only if delivery costs grow **[DECISION D19]**. KYC files and backups stay on AWS.
- KYC files and backups follow the hosting region decision (D2). af-south-1 (Cape Town) must be enabled on the AWS account before use.

8.7 Backup and recovery

Targets: recovery time (RTO) ≤ 4 hours and data loss (RPO) ≤ 1 hour for the whole platform.

What	How it's protected	Worst-case data loss	Kept for
PostgreSQL (users, listings, escrow, audit)	RDS Multi-AZ; automated backups with point-in-time recovery; daily AWS Backup snapshot copied to a second region and to the backup account	Minutes (point-in-time)	35 days point-in-time; monthly snapshots for 12 months
Audit logs	Monthly partitions exported to S3, then Glacier	None once exported	7 years
S3 files	Versioning; replication for originals, KYC and documents; Object Lock on documents and backups	Minutes (replication lag)	Per retention policy (D18)
Redis	Not backed up. Sessions and check-in tokens can be lost safely. Payment and	Sessions only	—

What	How it's protected	Worst-case data loss	Kept for
	escrow events always go to Postgres first (outbox)		
Typesense	Rebuilt from Postgres; optional snapshot for speed	None (rebuildable)	Latest snapshot
Infrastructure	Terraform in GitHub; secrets in Secrets Manager with replication	None	Git history
App signing keys	Play App Signing (Google holds the release key); upload key and iOS certificates in a team vault	None	Permanent

Restore drills

- Before launch, and then every quarter: restore the database and a sample of files into a test account, time it, and check the data.
- Record the time against the 4-hour RTO. A backup that has never been restored doesn't count as a backup.
- Once a year, practise rebuilding the whole environment from Terraform in the second region.

9. AI and data plan

RFC-003 calls the engine "superhuman" and promises 6,000× faster decisions, 99.2% fraud accuracy and 78% market prediction accuracy. Those figures have no data behind them yet, so they are removed from targets and marketing. The six-pod design stays as the long-term shape.

Capability	Release 1	Later	Data needed first
Fraud scoring	Weighted rules and perceptual hashing (Rust) + Isolation Forest (Python)	Graph ring detection (R2); supervised model (R3)	Reviewer labels from the R1 queue
Photo checks	Duplicate hash; reverse image search at review	ResNet stock/fake classifier (R2)	Labelled real vs fake photos
GPS anti-spoofing	Rule-based score	LSTM trajectory model (R2)	Real on-site check-ins
Search	Typesense keyword + geo	Semantic search with sentence-transformers (R2); Twi NLP (R3)	Query logs
Price guidance	Area price ranges from listings	XGBoost price model (R2)	Enough listings per area; threshold set by AI lead
Recommendations	Saved-search matches	LightFM hybrid (R2)	Views, saves, enquiries
Explanations	Plain-language reason list	SHAP values and counterfactuals (R2)	A trained model

9.1 Model rules that stay

- New model versions ship behind a 10% canary and replace the current one only if offline evaluation beats it by at least 1%.
- Model files versioned in S3 with SHA-256 checks.
- No model makes a final decision to ban a user or keep someone's money without human review in R1 and R2.

10. Success metrics

Targets kept from the source documents are marked "kept". Changed targets show the original in brackets.

10.1 Engineering

Metric	Target	Status
CI pipeline (lint, types, tests, Docker build)	Under 8 minutes	Kept
Hardcoded secrets in repo	Zero	Kept
API latency	p95 ≤ 300 ms at 1,000 concurrent users [was p99 ≤ 200 ms at 100K]	Amended for R1; 100K target moves to scale stage
Uptime	99.5% in R1, 99.9% from R3 [was 99.9% at launch]	Amended
GPS check-in latency	p99 < 800 ms at 200 concurrent	Kept
GPS false positives	Under 3% of real on-site users	Kept
Spoof detection	At least 97% on a QA suite of 20 mock-location apps	Kept
Fraud detection	Recall ≥ 95% and precision ≥ 90% on seeded fake listings [was 99.2% accuracy]	Amended
Payment sandbox	Zero failed end-to-end transactions	Kept
Backup and recovery	RTO ≤ 4 h, RPO ≤ 1 h, proven by a restore drill before launch and every quarter	Kept (drills added)
KYC file exposure	Zero publicly reachable KYC or document objects (automated weekly check)	New
Accessibility	WCAG 2.1 AA on all R1 screens	Kept (moved earlier)
Rust / Python parity	Identical hashes and scores (to 2 decimals) on the shared fixture set	New (STACK-001)
Photo hashing	p95 under 50 ms per 1080p photo in `afe_native`	New target (STACK-001)
Android start-up	Cold start under 2 s on a mid-range phone	New target (STACK-001)
App stability	Crash-free sessions ≥ 99.5% on both apps	New

10.2 Market (Ghana)

Metric	Target	Status
Verified agents and landlords in Accra and Kumasi	50 by end of private beta	Kept
Verified live listings	1,000 by end of private beta	Kept
Beta NPS	≥ 40 from a 50-person cohort	Kept
App store rating	≥ 4.2	Kept

Metric	Target	Status
Escrowed rent payments	Share of listed deals that settle through escrow — see Section 12.3	New
Revenue	Set from the model in Section 12 [was $25K MRR at week 34 across four countries]	Amended

11. Roadmap

ROADMAP.md puts launch at week 16. The Engineering Docs put a four-country launch at week 34. This plan launches Ghana at about week 28 and moves expansion after that.

Stage	Weeks	Goal	Exit criteria
0. Foundations	1–4	Repo, CI/CD, database, auth, design tokens, frontend API layer, Svelte 5 upgrade, Rust workspace + maturin build, Android and iOS project skeletons, S3 buckets and backups set up with Terraform	CI green; auth with MFA working in staging
1. Core build	5–14	Listings, verification, search, viewings + GPS check-in, escrow in sandbox, dashboards, admin queues; all R1 flows on web and Android; iOS core flows if staffed	A full rent journey works in staging with sandbox money
2. Hardening & legal	15–18	Security review, load test, Act 843 registration, escrow legal opinion, partner contract; first full restore drill; remaining iOS flows	No high-severity findings open; legal sign-off; restore drill within RTO
3. Private beta	19–24	Accra and Kumasi, invited agents and landlords, live payments with limits; Android closed testing, iOS TestFlight from about week 22	50 agents, 1,000 listings, NPS ≥ 40
4. Ghana launch	25–28	Public release on web, Google Play and the App Store; marketing; penetration test	Launch metrics in 10.1 met for four weeks
5. Grow (R2)	Months 8–12	ML models, reputation, basic intelligence, seller optimiser, Twi search	Models beat rules on held-out data
6. Expand (R3–R4)	Months 12–24	Loans, developer API, Nigeria and Kenya, then South Africa	Per-country legal sign-off before each launch

11.1 Build order inside Stage 1

1. Listings and photo upload (everything else hangs off a listing).
2. Verification score, duplicate detection and admin review queue.
3. Search and map.
4. Viewings and GPS check-in.
5. Escrow and compliance rules.
6. Reviews gated by check-in, then messaging and notifications.
7. Dashboards, using the Component Library and Quick Fixes conventions.

12. Business model and viability

Status of the numbers in this section

The **fee level and the subscription price** come from the existing documents (architecture.md, page-structure.md). Everything else — rents, volumes, costs, growth — is a placeholder.

The companion workbook, **Afe_Unit_Economics_Model.xlsx**, holds every figure as an editable input and works out contribution, break-even and runway. Replace the placeholders with the figures from the viability and monetization analysis; this section then follows the model, not the other way round.

12.1 How Afe makes money

Stream	Who pays	Rate	From
Escrow transaction fee	**[DECISION D6]** tenant, landlord, or split	3.5% of the escrowed amount, taken when escrow is released	Release 1
Premium agent subscription	Agents	Monthly; USD 50 equivalent in the source documents, to be priced in GHS **[DECISION D22]**	Release 1
Featured listings	Agents and landlords	Per listing, per period	Release 2
Lender referral fees	Lenders	Per funded loan	Release 3, once lenders sign
Market data and reports	Developers, lenders, agencies	Subscription	Release 4
Search and browsing	Tenants and buyers	Free, always	Release 1

Tenants pay nothing to search. Charging renters to see homes is what the informal market already does, and undercutting that is the point of the product.

One rate, stated once [DECISION D6]

The documents carry two different platform fees: **3.5%** in architecture.md and the pricing page, and **2.5%** in RFC-002 of the Engineering Docs. This statement uses 3.5% as the working figure, but the rate must be set once and referenced everywhere, because it is shown to users at payment time and written into the escrow record.

12.2 Unit economics

For one completed rental, revenue is the platform fee on the escrowed amount. Against it sit the payment gateway's cut of the whole sum moved, the identity and licence checks, notifications, storage, and reviewer time — including the disputes a minority of deals will raise.

- **The gateway fee is charged on the full escrowed amount, not on Afe's fee.** With six months' advance plus a deposit, the sum moved is large, so a gateway rate of 1.5–2.0% takes a large share of a 3.5% platform fee before any other cost.
- Every listing costs money to verify whether or not it ever rents. The model prices verification per transaction, so a low listing-to-let conversion rate makes the real economics worse than the model shows. Track that conversion from the first month.
- Subscriptions are the steadier line: almost no marginal cost, and revenue that does not depend on deals closing.

12.3 What has to be true

These are the assumptions the business rests on. Each needs a test during the private beta, not after launch.

Assumption	Why it matters	How to test it in beta
Deals settle through escrow rather than off-platform	This is the whole transaction line. Both sides can meet on Afe and then pay in cash to avoid the fee	Measure the share of viewings that end in an escrowed payment; ask agents why they skip it
Agents will pay a monthly subscription	The steadier revenue line	Offer paid tiers during beta instead of giving everything away; count who converts when the free period ends
Landlords accept the six-month cap	A landlord who insists on a year cannot transact on Afe at all	Count listings rejected at payment time for exceeding the cap
Verification cost per listing stays low	Reviewer time is the cost that scales worst with growth	Track minutes of review per listing and the share needing manual review
Fraud losses stay small	Refunds and disputes come straight out of contribution	Track disputed amounts as a share of escrowed volume

Leakage is the one to watch. A fee close to one month's rent gives both sides a strong reason to transact elsewhere, so the escrow has to be worth more to them than the fee costs: refund protection, a verified counterparty, and a receipt that stands up in a dispute.

12.4 Targets

The projections in ROADMAP.md (USD 20K a month by week 16, USD 250K by month 12) were written before the plan changed and have no working behind them. They are withdrawn. Fill these in from the model instead:

Stage	Escrowed deals / month	Subscribed agents	Revenue / month
Private beta (wk 19–24)	From the model	From the model	From the model
Ghana launch (wk 25–28)	From the model	From the model	From the model
Month 12	From the model	From the model	From the model
Month 24	From the model	From the model	From the model

[DECISION D23] Funding position and runway. The model turns cash on hand and monthly burn into a runway figure; the plan in Section 11 only holds if the runway covers it.

13. Team and budget

The source plans assume named leads for backend, frontend, mobile, AI, GPS, payments, DevOps, security and legal, growing from 6 to 20 people, with a year-one budget of about $780K. Those figures are USD estimates and were not checked against Ghana pay rates.

13.1 Smallest team that can ship Release 1

Role	Count	Notes
Founder / product lead	1	Owns scope, partners and legal process
Backend engineers (Python + Rust)	2	At least one comfortable in Rust; one also covers DevOps
Web engineer (SvelteKit)	1	
Android engineer (Kotlin)	1	Android ships first; owns Play Integrity and sensor work
iOS engineer (Swift)	1	Needs a Mac and an Apple Developer account; can join after Stage 0 **[DECISION D15]**
Product designer	0.5	Part-time or contract
Lawyer	Retainer	Act 220, Act 1047, Act 843, Act 987, terms of service
Trust & safety reviewer	1 from beta	Works the review and dispute queues

[DECISION] Confirm the real team size and budget. The roadmap in Section 11 and the model in Section 12 should be re-timed and re-costed against them.

14. Risks

Risk	Likelihood	Impact	Mitigation
Escrow treated as unlicensed money handling	Medium	High	Licensed partner; legal opinion before any live payment
GPS spoofing	High	High	Three layers; manual review band; audit trail
Not enough data for AI	High	Medium	Rules first; collect labels from day one
Scope creep from the 11 differentiators	High	High	Release gates in Section 11; nothing from 5.2 enters R1 without a decision
Agents avoid verification	Medium	High	Free verified badge, search ranking boost, faster payouts
Landlords refuse the six-month cap	High	Medium	Present compliance as protection; enforcement from April 2026 helps
Personal data breach	Medium	High	Row-level security, encryption at rest and in transit, least privilege, DPC registration
Tenancy law changes (pending Rent Bill)	Medium	Medium	Rules as config with dual approval
Established competitors (e.g. meqasa, Ownkey, Jiji)	High	Medium	Compete on escrow and verification, not listing count
High data costs for users	Medium	Medium	PWA, WebP, small bundles, SMS fallbacks
Small team burnout	Medium	High	Cut scope before cutting quality; monolith first
Two native apps drift apart	High	Medium	Generated API clients, shared design tokens, one acceptance script per feature
Rust skills hard to hire in Ghana	High	Medium	Keep Rust small and pure; Python reference code and parity tests let a Python engineer maintain it
Rust build slows CI or breaks images	Medium	Medium	Cached builds; wheel built once per commit; pinned toolchain
iOS app late	Medium	Medium	Android-first plan; iOS by launch or shortly after
Data loss or ransomware	Low	High	Point-in-time recovery, Object Lock, backups in a separate account and region, quarterly restore drills
KYC or document leak from storage	Medium	High	Private bucket, customer-managed key, short signed URLs, access logs, weekly public-exposure check

15. Decisions needed

D1. Confirm **Afe** as the name everywhere. Rename houseagent / HomeConnect / Hca prefixes in code, configs and component names (for example HcaTextInput → AfeTextInput).

D2. Hosting region: af-south-1 or eu-west-1.

D3. Container platform for R1: ECS Fargate or EKS.

D4. Ghana payment partner: Hubtel, Paystack or both.

D5. Escrow legal structure after counsel's opinion.

D6. Platform fee level (2.5% or 3.5%) and who pays it — tenant, landlord or split.

D7. Business model: which revenue streams run at launch, and the revenue targets in Section 12.4.

D8. Real team size and budget.

D9. Approve the Svelte 4 → 5 upgrade, MSW and Vitest install (implementation_plan.md).

D10. Staging gateway URLs, or MSW mocks for all services until the backend is up.

D11. MLflow or DVC.

D12. Twi strings in R1 or R2.

D13. Minimum iOS version: 16 or 17.

D14. Map provider on mobile: MapLibre (OSM) or Google Maps.

D15. If only one mobile engineer is hired: Android-only at launch, iOS after.

D16. Push notifications: FCM on both platforms, or APNs directly on iOS.

D17. iOS CI: GitHub Actions macOS runners or Xcode Cloud.

D18. Retention periods (with counsel, under Act 843): KYC files after account closure, photos from deleted listings, chat photos.

D19. Public photo delivery: CloudFront, or Cloudflare R2 if delivery costs grow.

D20. Second region for backups and replication (follows D2).

D21. Malware scanner for uploads: self-hosted ClamAV or a managed service.

D22. Premium agent subscription price in GHS, tested with real agents.

D23. Funding position and runway, entered into the unit economics model.

Appendix A — Amendment log

#	Topic	Before (source)	After
A1	Product name	House Agent; HomeConnect Africa; Sheltr recommended	Afe
A2	Launch	Four countries at once (Eng Docs); five in README	Ghana first; NG and KE in R3; ZA in R4; TZ backlog
A3	Backend shape	8 → 11 microservices (12 listed)	Modular monolith, 12 modules
A4	Web framework	React 18 / Vue / Next.js in different files	SvelteKit 2 + Svelte 5
A5	Search engine	Elasticsearch	Typesense
A6	Messaging backbone	Kafka + gRPC	Celery, Redis Streams, in-process calls
A7	Admin UI	Django Admin	Admin role in SvelteKit
A8	Postgres version	15 and 16	16
A9	Ghana rent law	"Rent Act (2021)"	Rent Act, 1963 (Act 220); new Rent Bill still pending
A10	Ghana agents	Not covered	Real Estate Agency Act, 2020 (Act 1047) licence check
A11	Ghana mobile money	MTN MoMo + Vodafone Cash	MTN MoMo, Telecel Cash (renamed), AT Money
A12	Nigeria data law	NDPR	Nigeria Data Protection Act, 2023
A13	Ghana languages	Twi missing from the 4-language launch list	English plus Twi, Ga, Ewe
A14	Escrow licensing	"Gateway wallet avoids licensing" assumed	Legal opinion required (Act 987)
A15	PCI-DSS	Audit per country	Hosted checkout; minimal scope; not country-based
A16	AI claims	Superhuman, 6,000×, 99.2% accuracy, 78% prediction	Removed; precision/recall targets
A17	AI at launch	Six model pods in Phase 2	Rules + hashing + Isolation Forest; models in R2
A18	Verification weights	ML anomaly 10% from day one	0% until a model exists
A19	Ownership proof	Title deed	Title, indenture, site plan or landlord authorisation
A20	Rooted devices	Rejected at check-in	Sent to manual review
A21	Scale targets	100K concurrent, p99 200 ms at launch	1,000 concurrent, p95 300 ms in R1
A22	Revenue target	$25K MRR at week 34	Pending business model decision
A23	Mortgage calculator	Phase 3, "CRITICAL"	R1, client-side
A24	Loan matching	Phase 2, "CRITICAL"	R3, after lender partners sign

#	Topic	Before (source)	After
A25	Timeline	16 weeks (ROADMAP) vs 34 weeks (Eng Docs)	Ghana launch around week 28
A26	Schema	"26 tables" (about 15 defined)	Missing tables listed in 8.5
A27	Dependency pins	FastAPI 0.104, Pydantic 2.5, Postgres 15 image (2023)	Update to current stable versions at Stage 0
A28	Mobile framework	Flutter (one codebase)	Native Kotlin (Android) and Swift (iOS)
A29	iOS minimum	iOS 14	iOS 16 recommended (D13)
A30	CPU-heavy backend code	Python; "move to Go" later	Rust via PyO3 now, for pure functions only
A31	Python runtime	python:3.11-slim	python:3.12-slim with a Rust build stage
A32	Mobile components	Flutter prompts CL-080–082, CL-090–094 (Hca prefix)	Compose + SwiftUI prompts (Afe prefix) in STACK-001
A33	Mobile testing	flutter test, WidgetTester, integration_test	JUnit, Compose UI test, XCTest, XCUITest
A34	Mobile team	1 Flutter engineer	1 Android + 1 iOS engineer
A35	Mobile release order	Both platforms together	Android beta first; iOS by launch
A36	File storage	One S3 store for images, documents and backups	Five buckets split by sensitivity (8.6)
A37	Image records	property_images.url (full URL)	S3 keys + stored_files table
A38	Photo metadata	EXIF stored and file kept as uploaded	GPS/date read into the database, then stripped from the file
A39	Backups	30-day RDS snapshots; DR region us-east-1	Point-in-time recovery; copies to a second region and a separate account; region per D2/D20
A40	Event durability	Events could sit only in Redis	Outbox table in Postgres for payment and escrow events
A42	Platform fee	3.5% (architecture.md, pricing page) vs 2.5% (RFC-002)	One rate, set by D6; 3.5% used as the working figure
A43	Revenue projections	$20K/month by week 16; $250K by month 12 (ROADMAP.md)	Withdrawn; replaced by the model in Section 12.4
A44	Pricing page	Free / commission / premium tiers with no costings	Same tiers, now backed by unit economics
A41	Recovery testing	"Tested before launch"	Restore drill before launch and every quarter

Appendix B — Source document map

File(s)	Role now	Action
README.md, PACKAGE-INDEX.md, QUICKSTART.md	Package overview and setup	Rename to Afe; update stack and countries
SKILL.md	Claude skill for the project	Rename; remove Django Admin; set Ghana first
ROADMAP.md	Old timeline and revenue projections	Superseded by Sections 11 and 12.4
naming-options.md	Name search	Superseded (name is Afe)
architecture.md, architecture-country.md	Near-duplicate Nigeria architecture docs; now include storage and backup	Merge into one; regenerate for Ghana with generate_architecture_doc.py
framework-comparison.md	Backend choice	Canonical; basis for the monolith
database-schema.md, init_database.py	Base schema; stored_files added, image URLs replaced by S3 keys	Canonical; extend per 8.5
api-design.md	API patterns	Canonical
fraud-detection.md	Fraud algorithms	Canonical; apply weight change in 6.1
page-structure.md	Sitemap	Reference; align routes with Other Pages prompts (/properties vs /search)
HomeConnect_Africa_Engineering_Docs_v2.docx	OKRs, RFCs, review checklists, PRs	Superseded in part by Sections 6–11; RFC and checklist detail still useful
HomeConnect_Africa_Implementation_Plan.docx	15 frontend-gap work items	Re-phased per 5.2
Afe_Stack_Amendment_STACK-001.docx	Rust and native mobile detail	Canonical for everything it covers
Afe_Unit_Economics_Model.xlsx	Unit economics, projection and runway	Fill in the Inputs sheet from the viability analysis; Section 12 follows it
HomeConnect_ComponentLibrary.docx, _Dashboard_Prompts.docx, _OtherPages_Prompts.docx	Build prompts	Use with Quick Fixes and STACK-001 applied; mobile prompts replaced by STACK-001; trim to R1 cards
HomeConnect_QuickFixes.docx	Amendments to the prompt docs	Canonical, except Flutter parts of Fix 01, Fix 02 and Fix 10 (replaced by STACK-001)
HomeConnect_API_InfraPrompt.docx (+ _extracted.txt)	Frontend API layer spec	Canonical; delete the .txt duplicate
implementation_plan.md	Plan for the frontend API layer	Proceed once decision 9 is made
Gray CSM UI Design System#2.md	Design system written for a separate Next.js/shadcn product	Use only its visual rules and its HomeConnect modernisation

File(s)	Role now	Action
		phases; ignore Next.js/shadcn specifics
Dockerfile, docker-compose.yml, kubernetes.yaml, requirements.txt	Infrastructure baseline	Rename houseagent → afe; update versions; pin Grafana/Prometheus instead of :latest. docker-compose now includes MinIO as local S3

Sources checked for Section 2 and 7 (September 2026): reporting on Rent Control Department enforcement of the Act 220 advance-rent cap from 1 April 2026 (ModernGhana, May 2026); tenancy guides noting the Rent Bill pending since 2023 and the Act 1047 licensing regime (Ownkey, 2026).